

Homotopy.

Def. Let $f, g: X \rightarrow Y$ be continuous maps.

We said that f and g are Homotopic if there exists a Continuous map

$$F: X \times I \rightarrow Y \text{ such that } \begin{cases} F(x, 0) = f(x), & \forall x \in X. \\ F(x, 1) = g(x), & \forall x \in X. \end{cases}$$

This F is called a Homotopy, and in such a case, denote $f \simeq g$.

Thm. Let $C(X, Y) \stackrel{\text{def}}{=} \{f: X \rightarrow Y \mid f \text{ is continuous}\}$,

Define a relation on $C(X, Y)$ as: $f \sim g \Leftrightarrow f \simeq g$.

Then, the Homotopy is equivalence relation.

Proof. Reflexivity: Given $f \in C(X, Y)$, define

$$F: X \times I \rightarrow Y: (x, t) \mapsto f(x)$$

Then, since $F^{-1}[V] = f^{-1}[V] \times I$, F is continuous such that $F(x, 0) = F(x, 1) = f(x)$.

Symmetric. Let $f \simeq g$ via $F: X \times I \rightarrow Y$. Define $G: X \times I \rightarrow Y: (x, t) \mapsto F(x, 1-t)$

Then, $G(x, 0) = F(x, 1) = g(x)$ and $G(x, 1) = F(x, 0) = f(x)$.

Transitivity. Let $f \simeq g$ via F and $g \simeq h$ via G . Define

$$H: X \times I \rightarrow Y: (x, t) \mapsto \begin{cases} F(x, 2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ G(x, 2t-1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$